

Cosmological Theory & CosmoGravity

The summary of Cosmological Theory presented below focuses on the mathematical relations that are used in the code of the UNIVERS simulations of CosmoGravity. It nonetheless begins with a brief presentation of general relativity and of its link with “cosmology”, which is the study of the Universe as a whole. And towards the end a transition is proposed (a physically rather rapid structuring) that leads to the TRAJECTOIRES simulation, which is the study in general relativity of the local gravitational environment of a single massive object (star, planet, ... black hole)

1 General relativity

General relativity is a **relativistic geometric theory of gravitation (RGTG)** — that is, a theory that “connects” the geometry of the container (space-time) to its content (matter-energy) and which, locally, reduces to special relativity¹.

2 The Cosmological Principle

The cosmological principle postulates that **the universe is homogeneous and isotropic** — that is, the same everywhere (3D space) and in every direction. The only possible changes are therefore with time: a time that is the same everywhere since nothing “moves”². This is indeed the hypothesis of a very strong and strange symmetry³.

When Einstein used it in 1917, it was still hazardous to assume that the observed sky, then at small distances, was uniform on much larger scales. And this “Cosmological Principle” was largely motivated by the fact that its extreme symmetry made it possible, together with general relativity, to derive “analytic” solutions at a time when “numerical” computation and large computing power still lay on a very distant horizon.

But, progressively, observations of celestial objects, then those of the cosmic microwave background (CMB)⁴, have reinforced the plausibility of this principle, at least at large distances (today at $d > 10^{24}$ m). And this CMB shows us a content in the state of an opaque plasma when, about 380,000 years after the Big Bang, it becomes a neutral, transparent gas that is remarkably homogeneous, since, in mass density ρ , its inhomogeneities are tiny: $\delta\rho/\rho \approx 10^{-5}$.

¹Special relativity (Einstein, 1905) already “connected”, on the one hand, space and time (Minkowski space-time) and, on the other hand, matter and energy through the equivalence $E = mc^2$

²changes “location”

³At our own scale we are in fact much more used to thinking of a landscape, a structure, that changes from one point of space to another but very little or very slowly with time, than the opposite

⁴CMB (or CMBR) for Cosmic Microwave Background (Radiation)

3 The Friedmann-Lemaître-Robertson-Walker metric

Any **RGTG** combined with the cosmological principle implies that the symmetry of the content (matter-energy) is the same as that of the container (space-time), and that the most general form of its “*metric*” (the square ds^2 of the 4D space-time length element) is that of Friedmann-Lemaître-Robertson-Walker (**FLRW**)⁵ :

$$ds^2 = -R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\varphi^2 \right] + c^2 dt^2 \quad \text{with} \quad k = -1, 0 \text{ or } +1 \quad (1)$$

where r is a radial coordinate and where $R(t)$ is the “*scale factor*”, $R(t)$ being a real, well-defined, positive function of the *cosmic time* variable t that multiplies the distance between the fixed points of 3D space. The 3 possibilities for k define the 3 types of simply-connected spatial topology compatible with the assumptions: S^3 ([hyper]-spherical space), (Euclidean space E^3), H^3 ([hyper]-hyperbolic space)⁶.

The time t is the “*cosmic time*”. It is orthogonal (independent of the 3 spatial coordinates). It is the same for all observers “at rest”⁷ (r, θ and φ constant), often called “*comoving*” (somewhat improperly, since they do not “move”).

The expansion rate $H(t)$ is defined as the logarithmic derivative of $R(t)$: $H(t) \stackrel{\text{def}}{=} \dot{R}(t)/R(t)$. Its present value is written $H_0 = H(t_0)$ and its dimension is therefore the inverse of a time. For historical reasons tied to the method of measuring distances it is often expressed in $\text{km s}^{-1} \text{ Mpc}^{-1}$ ⁸.

By following the trajectory of the photons (for which $ds = 0$) between emission and reception, one deduces that if a source emits two light signals, the first at time t_e and the second at time $t_e + dt_e$, an observer will receive them at times t_0 and $t_0 + dt_0$ with:

$$\frac{dt_0}{dt_e} = \frac{R(t_0)}{R(t_e)} \quad (2)$$

For very small periods T , wavelengths λ and high frequencies ν of light, this can be written:

$$\frac{R(t_0)}{R(t_e)} \approx \frac{T_0}{T_e} = \frac{\lambda_0}{\lambda_e} = \frac{\nu_e}{\nu_0} \stackrel{\text{def}}{=} 1 + z \quad (3)$$

with z denoting, by default, the cosmological redshift.

4 Spectral shifts

There are 3 main spectral shifts: cosmological z_c , Doppler z_D and gravitational z_g . z_D (the Doppler effect) is due to a genuine motion of the source and/or the observer, and z_g to local

⁵established in an increasingly general way between 1922 and 1935 by the 4 authors

⁶The cosmological principle implies that the curvature (positive, negative or zero) of 3D space is, in each of the 3 cases, the same everywhere

⁷This “cosmic time” is approximately the time on one’s watch (or smartphone) as long as the observer neither moves at relativistic speeds nor ventures close to the Schwarzschild radius of a mass

⁸Mpc: Megaparsec, 1 Mpc = 10^6 pc and 1 pc (parsec) $\stackrel{\text{def}}{=} 3.0856775814672 \cdot 10^{16}$ m

gravitation. Generally, in the observation of a distant object the three causes of spectral shift act in succession to produce the observed spectral shift: as it moves away from a massive object the light undergoes a z_g , then on a larger scale its local motion generates a shift z_D , and finally at very large distances the scale factor of space produces a cosmological spectral shift z_c . This succession in time then allows one to write that the observed spectral shift z is the result of the chronological sequence (gravitation, velocity, cosmology) of the 3 causes according to $1 + z_g$, then $1 + z_D$ and finally $1 + z_c$, i.e. the relation:

$$1 + z = (1 + z_c) \cdot (1 + z_D) \cdot (1 + z_g) \quad (4)$$

Since for distant objects (those affected by the variation of the scale factor) one is most often in the case where $z_c \gg z_D \gg z_g$, the observed z differs little from z_c . In the remainder of this text we shall, by default, identify z with z_c . One may note that this is in fact formally exact under the cosmological principle.

5 The Universe and general relativity

Einstein's general relativity, published in 1915-1916, was first applied to local gravitation (Schwarzschild, 1916), which is the domain of the "Trajectoires" simulations of CosmoGravity. The transition from our (almost) perfectly homogeneous early universe to the extremely structured one of today will be described towards the end of this document.

Einstein was the first, in 1917⁹ to apply his theory to the Universe and, in order to obtain a static model¹⁰, he added to his equation the "cosmological constant", stating: "*We may indeed add, on the left-hand side of the field equation, the fundamental tensor $g_{\mu\nu}$ multiplied by a provisionally unknown **universal constant** $-\lambda$, without destroying the general covariance; we substitute for the field equation*":

$$\frac{1}{2}g_{\mu\nu}\mathcal{R} - \mathcal{R}_{\mu\nu} - \lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\mathcal{T}_{\mu\nu} \quad (5)$$

It is worth noting that with the addition of the term $\lambda g_{\mu\nu}$ the left-hand side of the equation becomes the most general tensor (built from the $g_{\mu\nu}$) whose covariant derivative is zero.

This lower-case letter λ has become the upper-case Λ in usual notation¹¹.

The $g_{\mu\nu}$ are the coefficients of the metric. Because of its symmetry, only the diagonal terms ($\mu = \nu$) of the FLRW metric (1) are non-zero:

$$g_{11} = -\frac{R^2(t)}{1 - kr^2}, \quad g_{22} = -R^2(t)r^2, \quad g_{33} = -R^2(t)r^2 \sin^2 \theta, \quad g_{44} = c^2 \quad (6)$$

6 Friedmann-Lemaître universe models

With general relativity as the **RGTG**, introducing the coefficients ($g_{\mu\nu}$) of dr^2 , $d\theta^2$, $d\varphi^2$ and dt^2 of the FLRW metric into Einstein's equation transforms it into 2 differential equations,

⁹Einstein, 1917, Sitzungsberichte der Königlich PreuBischen Akademie der Wissenschaften (Berlin), Seite 142-152

¹⁰neither expansion nor contraction of 3D space

¹¹ λ often denoting the "wavelength" in physics

the Friedmann-Lemaître equations FL1 and FL2, for the scale factor $\mathbf{R}(\mathbf{t})$, in which $p(t)$ and $\rho(t)$ (pressure and mass density) are the only two physical parameters describing the content: p and ρ which are, according to the cosmological principle, spatially homogeneous (hence not functions of r , θ or φ) and isotropic (hence scalar, even for the pressure)¹². $\dot{R}(t)$ and $\ddot{R}(t)$ denote the derivatives dR/dt and d^2R/dt^2)

$$-\frac{k}{R^2} - \frac{\dot{R}^2}{c^2 R^2} - \frac{2\ddot{R}}{Rc^2} + \Lambda = \frac{8\pi G p}{c^4} \quad (\text{FL1}) \quad \text{and} \quad \frac{k}{R^2} + \frac{\dot{R}^2}{c^2 R^2} - \frac{\Lambda}{3} = \frac{8\pi G \rho}{3c^2} \quad (\text{FL2}) \quad (7)$$

FL3 is deduced from FL1 and FL2:

$$\frac{d(\rho R^3)}{dR} + 3p \frac{R^2}{c^2} = 0 \quad (\text{FL3}) \quad (8)$$

And FL2 can also be written:

$$\dot{R}^2 = H_o^2 R_o^2 \left[\Omega_{r_o} \frac{R_o^2}{R^2} + \Omega_{m_o} \frac{R_o}{R} + \Omega_{\Lambda_o} \frac{R^2}{R_o^2} + \Omega_{k_o} \right] \quad (9)$$

with

$$\Omega_r(\mathbf{t}) \stackrel{\text{def}}{=} \frac{8\pi G \rho_r(\mathbf{t})}{3H^2(\mathbf{t})}, \quad \Omega_m(\mathbf{t}) \stackrel{\text{def}}{=} \frac{8\pi G \rho_m(\mathbf{t})}{3H^2(\mathbf{t})}, \quad \Omega_\Lambda(\mathbf{t}) \stackrel{\text{def}}{=} \frac{\Lambda c^2}{3H^2(\mathbf{t})} \quad \text{and} \quad \Omega_k(\mathbf{t}) \stackrel{\text{def}}{=} -\frac{kc^2}{R^2(\mathbf{t})H^2(\mathbf{t})} \quad (10)$$

G is the constant in Newton's law of universal gravitation between two masses m_1 and m_2 separated by a distance d ¹³.

$\Omega_r(t)$ is the radiation density parameter (light or ultra-relativistic particles, which have the same equation of state $p = \frac{1}{3}\rho c^2$) (cf. §7 Primordial neutrinos).

$\Omega_m(t)$ is the matter density parameter (including non-baryonic dark matter)

$\Omega_\Lambda(t)$ is the density parameter of Λ . It is also called the reduced cosmological constant (but $\Omega_\Lambda(t)$ is generally not a constant).

$\Omega_k(t)$ is the curvature density parameter (or reduced curvature). It depends on k , which represents the curvature of the universe. If:

- $k = 1$: [hyper]-spherical (3D) space
- $k = 0$: Euclidean (3D) space (in this case $\Omega_k(t) = 0 \quad \forall t$)
- $k = -1$: [hyper]-hyperbolic (3D) space.

Ω_{r0} is the best-known density parameter, because the light-energy density ρ_{r0} is essentially that of the cosmic microwave background (CMB). One may plausibly add to it that of the primordial neutrinos, not yet detected, which in the simplest hypothesis have an energy density equal to $\sim 69\%$ of that of the CMB (cf. next §).

From FL2 one deduces:

$$\Omega_r(\mathbf{t}) + \Omega_m(\mathbf{t}) + \Omega_\Lambda(\mathbf{t}) + \Omega_k(\mathbf{t}) = 1 \quad \forall \mathbf{t} \quad (11)$$

¹²the fluids of the content are therefore *perfect fluids (without viscosity)*

¹³ $F = -Gm_1m_2/d^2$

It is therefore enough to measure three of the four Ω_i at a given instant to determine the fourth¹⁴. Since today Ω_{r0} is well known and, moreover, smaller than 10^{-4} , knowledge of our universe model is essentially tied to the measurements of Ω_{m0} and $\Omega_{\Lambda0}$.

The interactive graph thus makes it possible to simulate different universes with a simple click on the $\Omega_{m0} - \Omega_{\Lambda0}$ diagram (the values of the other parameters are, by default, those of the Λ -CDM model derived from the 2015 results of ESA's Planck mission). Note that this approximation is only valid for the epochs where Ω_{m0} and $\Omega_{\Lambda0}$ are dominant. This is why it is more accurate to modify the present numerical values of the cosmological parameters. For Ω_{r0} , since the CMB has a thermal black-body spectrum, it is the temperature T_0 that must be chosen. An option also allows one to choose (force) $\Omega_k = 0 = \Omega_{k0}$. Finally, the code can also take primordial neutrinos into account (next section) and reach a reasonable accuracy beyond the first second for the observed universe.

The interactive graph shows the separatrices between different kinds of $a(t)$ curves, with or without Big Bang or Big Crunch, ...

Choosing a universe model very close to one (a fortiori two) separatrix(es) may in some cases cause the code to fail.

The mass density ρ_r ($\rho_r = u_r/c^2$ with u_r the energy density) of a black body depends only on its temperature T :

$$\rho_r = \frac{4\sigma T^4}{c^3} \quad \text{with} \quad \sigma = \frac{2\pi^5 k^4}{15h^3 c^2} \quad (12)$$

By default the **fundamental constants** have the values of our universe but they remain freely modifiable in the simulations (in the multiverse hypothesis they could indeed be different):

- **Boltzmann k** = $1.380649 \cdot 10^{-23} \text{ m}^2.\text{kg}.\text{s}^{-2}.\text{K}^{-1}$ (CGPM 2018)
- **Planck h** = $6.62607015 \cdot 10^{-34} \text{ J} \cdot \text{s}$ or $\text{kg m}^2 \text{ s}^{-1}$, (CGPM 2018)
- **universal gravitation G** = $6.67430(15) \cdot 10^{-11} \text{ m}^3.\text{kg}^{-1}.\text{s}^{-2}$ (CODATA 2018)
- **speed of light in vacuum c** = $299792458 \text{ m} \cdot \text{s}^{-1}$ (CGPM 1983)

We define the reduced coordinates: $\mathbf{a} \stackrel{\text{def}}{=} \mathbf{R}(\mathbf{t})/\mathbf{R}(\mathbf{t}_0)$ (consequently $a = (1+z)^{-1}$ or $z = (1-a)/a$ if z is purely cosmological) and $\tau \stackrel{\text{def}}{=} \mathbf{H}_0(\mathbf{t} - \mathbf{t}_0)$; the first and second derivatives of $a(\tau)$ are written:

$$\frac{d\mathbf{a}}{d\tau} = \left[\frac{\Omega_{r0}}{a^2} + \frac{\Omega_{m0}}{a} + \Omega_{\Lambda0} a^2 + \Omega_{k0} \right]^{\frac{1}{2}} \quad (13)$$

$$\frac{d^2\mathbf{a}}{d\tau^2} = -\frac{\Omega_{r0}}{a^3} - \frac{1}{2} \frac{\Omega_{m0}}{a^2} + \Omega_{\Lambda0} a \quad (14)$$

with the initial conditions: $a(0) = (da/d\tau)(0) = 1$.

¹⁴that was obviously the purpose of this parametrisation

Thus the data H_0 , Ω_{r0} , Ω_{m0} , and $\Omega_{\Lambda0}$ that are deduced from observations make it possible to solve the above differential equation and hence to know $a(\tau)$ for all τ (and from there $a(t)$ for all t), and therefore to plot its graph (which is the first action of the simulation part).

One can of course question the measured values of the parameters of our universe, or simply imagine different universes. Depending on the chosen values of the Ω_i , one then obtains different models, with or without singularity(ies). If $H_0 > 0$:

- Universe with a Big Bang and no Big Crunch
- Universe with a Big Bang and a Big Crunch
- universe without a Big Bang but with a Big Crunch
- Universe without a Big Bang or a Big Crunch

In our present universe the radiation density parameter is very small ($\Omega_{r0} < 10^{-4}$), while Ω_{m0} and $\Omega_{\Lambda0}$ are close to 1/3 and 2/3. Under these conditions it is essentially Ω_{m0} and $\Omega_{\Lambda0}$ that determine the type of universe: the separatrices in the $\{\Omega_{m0}, \Omega_{\Lambda0}\}$ plane are shown on the interactive diagram to the left of the simulation window.

CosmoGravity makes it possible to simulate a very wide variety of UNIVERS, many of which are incompatible with our presence in the universe. For example, those without a Big Bang cannot explain the nucleosynthesis of the light elements or the cosmic microwave background, and those with too short a duration seem incompatible with the billions of years required for the evolution of matter¹⁵ and of life¹⁶.

7 Primordial neutrinos

Although not yet detectable, because of their very low energies and despite the still-open question of their very small (but non-zero?) mass, they are probably as numerous as the photons and have a Planck (black-body) spectrum but at a lower temperature, owing to their earlier decoupling from matter. Expressed in terms of density, combining them with the CMB photons amounts¹⁷ to multiplying the photon parameter Ω_r by 1.6913. CosmoGravity Univers lets one choose to take into account the CMB alone, the CMB and the neutrinos, or neither: by default the CMB and the neutrinos are included in the calculation.

8 Computing durations and ages

Still following the trajectory of the photons, one derives the relations (between distances, times, spectral shifts, apparent diameters, ...) that are used in CosmoGravity (in particular in the toolbox of the “Cosmological Calculator” window). These expressions often use the

¹⁵stardust

¹⁶these *anthropic* constraints are far weaker in the TRAJECTOIRES simulations, since our present universe contains an extreme variety of different objects

¹⁷approximation with an *effective* number of neutrino flavours of 3.045 instead of 3, to compensate for the incomplete decoupling of the neutrinos during electron-positron annihilation (around $t \approx 1s$)

function $E(x)$, very convenient for simplifying the writing of the relations. It is defined arbitrarily as:

$$\mathbf{E}(\mathbf{x}) \stackrel{\text{def}}{=} \Omega_{\text{ro}}(1 + \mathbf{x})^4 + \Omega_{\text{mo}}(1 + \mathbf{x})^3 + (1 - \Omega_{\text{mo}} - \Omega_{\text{ro}} - \Omega_{\Lambda\text{o}})(1 + \mathbf{x})^2 + \Omega_{\Lambda\text{o}} \quad (15)$$

and one then obtains:

$$\frac{\mathbf{H}(\mathbf{z})}{\mathbf{H}_\text{o}} = \mathbf{E}^{\frac{1}{2}}(\mathbf{z}) = \left[\Omega_{\text{ro}}(1 + z)^4 + \Omega_{\text{mo}}(1 + z)^3 + (1 - \Omega_{\text{mo}} - \Omega_{\text{ro}} - \Omega_{\Lambda\text{o}})(1 + z)^2 + \Omega_{\Lambda\text{o}} \right]^{\frac{1}{2}} \quad (16)$$

The value of this simple expression $H(z)/H_0 = E^{1/2}(z)$ lies in the simplicity of the link between dt and dz along a line of sight, and in the fact that, unlike t , z **is an observable**.

From $z = R_0/R - 1$ one deduces $dz = -(1+z) H(t)dt$ and therefore: $d\mathbf{t} = -\mathbf{H}_0^{-1}(1 + \mathbf{z})^{-1}\mathbf{E}^{-1/2}(\mathbf{z})d\mathbf{z}$

Integrating this relation gives the durations as a function of the shifts:

$$\mathbf{t}_2 - \mathbf{t}_1 = \frac{1}{\mathbf{H}_\text{o}} \int_{\mathbf{z}_2}^{\mathbf{z}_1} (1 + \mathbf{x})^{-1} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \quad (17)$$

From this one deduces the expression for the age (t_o) of an FL universe with a Big Bang as a function of its parameters H_o and $\Omega_{i\text{o}}$, and that of the age of this universe at the emission of light received today with a spectral shift z :

$$\mathbf{t}_\text{o} = \frac{1}{\mathbf{H}_\text{o}} \int_0^\infty (1 + \mathbf{x})^{-1} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} ; \quad \mathbf{t}_\text{e} = \frac{1}{\mathbf{H}_\text{o}} \int_z^\infty (1 + \mathbf{x})^{-1} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \quad (18)$$

CosmoGravity also allows the inverse calculation: z as a function of t .

Note on the cosmological spectral shift and calculations of the future:

For a universe whose $a(t)$ increases continuously from 0 to infinity, like our standard Λ CDM model, $z(t_0) = 0$ at the present instant, $z \rightarrow +\infty$ the further we observe into the past, and ... $z \rightarrow -1$ in the future. In short: $-1 < z < \infty$. Obviously, these negative z (for models with increasing $a(t)$) are not observable today, but they make it possible to ... compute the future (cf. §7 and 8), as for example for the present distance (and hence the measurable z) of the most distant objects to which we could today send a message that could be received in a finite time: our present “*cosmological event horizon*”.

9 Computing distances

One obtains the exact expression of the “metric distance” d_m by “integrating” the trajectory of a photon from its emission at time t_e and coordinate r to its reception at time t_o at $r = 0$ with a cosmological spectral shift z .

Depending on whether the spatial curvature is negative (d_{m-}), zero (d_{m0}) or positive (d_{m+})

$$\mathbf{d}_{m-} = \frac{\mathbf{c}}{\mathbf{H}_\text{o} \mid \Omega_{\text{k}\text{o}} \mid^{\frac{1}{2}}} \sinh \left\{ \mid \Omega_{\text{k}\text{o}} \mid^{\frac{1}{2}} \int_0^{z_c} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \right\} \quad (19)$$

$$\mathbf{d}_{m0} = \frac{\mathbf{c}}{\mathbf{H}_\text{o}} \int_0^{z_c} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \quad (20)$$

$$\mathbf{d}_{m+} = \frac{\mathbf{c}}{\mathbf{H}_\text{o} \mid \Omega_{\text{k}\text{o}} \mid^{\frac{1}{2}}} \sin \left\{ \mid \Omega_{\text{k}\text{o}} \mid^{\frac{1}{2}} \int_0^{z_c} \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \right\} \quad (21)$$

CosmoGravity also allows the inverse calculation: z as a function of d

10 Computing horizons

In the main Univers window of CosmoGravity the distances and spectral shifts of these horizons are computed for the present time t_0 (d_{p0} and z_{p0} for the particle horizon and d_{e0} and z_{e0} for the event horizon).

10.1 Cosmological particle horizon

The cosmological particle horizon is the largest present metric distance d_{p0} from which we can receive a signal today (t_0). These largest distances correspond to an infinite z and are therefore expressed:

$$d_{p0-} = \frac{c}{H_0 |\Omega_{k0}|^{\frac{1}{2}}} \sinh \left\{ |\Omega_{k0}|^{\frac{1}{2}} \int_0^\infty E^{-\frac{1}{2}}(x) dx \right\} \quad (22)$$

$$d_{p0} = \frac{c}{H_0} \int_0^\infty E^{-\frac{1}{2}}(x) dx \quad (23)$$

$$d_{p0+} = \frac{c}{H_0 |\Omega_{k0}|^{\frac{1}{2}}} \sin \left\{ |\Omega_{k0}|^{\frac{1}{2}} \int_0^\infty E^{-\frac{1}{2}}(x) dx \right\} \quad (24)$$

10.2 Cosmological event horizon

The cosmological event horizon is the largest distance d_e to which we can today (t_0) send a signal that will be received in a finite time. It is also the largest distance of an object from which we can receive, in a finite time, a signal emitted today. CosmoGravity computes this distance on the main page for the present time t_0 . The expressions for d_{e0-} , d_{e0} and d_{e0+} are similar to those for d_{p0-} , d_{p0} and d_{p0+} , replacing the integration bounds 0 and ∞ by -1 and 0 for universes with a Big Bang and without a Big Crunch.

11 Computing the drift of the cosmological spectral shift

Over time the observed cosmological spectral shift of a fixed object varies with the scale factor $a(t)$. Let z_S denote the shift of this distant source observed today (at t_0). Measuring this change appears difficult to achieve, because the drift over a few years or a few tens of years is very small and is, moreover, contaminated by the other causes of spectral shift (essentially the Doppler-Fizeau shift due to a genuine motion). But with the performance of the next very large surveys, the signal-to-noise ratio will be able to be considerably improved by the increase in the numbers of measurements and of measurable objects.

Assuming the shift is purely cosmological (Univers Theory, equation 3):

$$z(t_0) = \frac{a(t_0)}{a(t_e)} - 1 \quad (25)$$

and following Balbi and Quercellini, 2007¹⁸

$$z(t_0 + \Delta t_0) = \frac{a(t_0 + \Delta t_0)}{a(t_e + \Delta t_e)} - 1 \quad (26)$$

¹⁸Balbi and Quercellini, 2007, “The time evolution of cosmological redshift as a test of dark energy” MNRAS 382, 1623, <https://articles.adsabs.harvard.edu/pdf/2007MNRAS.382.1623B>

$$\Delta z = z(t_0 + \Delta t_0) - z(t_0) = \frac{a(t_0 + \Delta t_0)}{a(t_e + \Delta t_e)} - \frac{a(t_0)}{a(t_e)} \quad (27)$$

To first order in $\Delta t_0/t - 0$ (“*humanly speaking*” $\Delta t_0/t_0 \ll 1$)

$$\Delta z \simeq \Delta t_0 \left[\frac{\dot{a}(t_0) - \dot{a}(t_e)}{a(t_e)} \right] \quad (28)$$

Since

$$H(z) \stackrel{\text{def}}{=} \dot{a}(z)/a(z) \quad (29)$$

the spectral-shift drift can be expressed ($\Delta t_0 \ll t_0$):

$$\frac{\Delta z}{\Delta t_0} \simeq [H_0 \cdot (1 + z_S) - H(z_S)] = H_0 \cdot [1 + z_S - E^{1/2}(z_S)] \quad (30)$$

The spectral-shift drift is therefore linked to the function $H(z)$ (cf. equation 16), which depends on the cosmological parameters H_0 and Ω_{i0} . Measuring this drift of z would be an independent method for better characterising the various cosmic fluids and, in particular, the time evolution of the density parameters and, from there, their equations of state. The dimension of $\Delta z/\Delta t_0$ is the inverse of a time. Its practical unit is yr^{-1} .

Remark: There is no shift drift for a z_S such that $H(z_S) = (1 + z_S)H_0$.

12 Computing apparent diameters

Because of the necessarily radial trajectory of the photons received in an isotropic space-time, the apparent diameter of an object (or the angular separation between two sources of the same z) is defined at the moment of emission, i.e. at the moment when the distance of the source was $(1 + z)$ times smaller than at the moment of observation. More mathematically, the metric of the surface $r = r$ and $t = t_e$ is that of a Euclidean 2-sphere of radius $R_e r = R_0 r/(1 + z)$, and the Euclidean relation between the linear diameter D_e , the metric distance d_m and the apparent diameter ϕ_0 can therefore be used:

$$\phi_0 = \frac{\mathbf{D_e(1 + z)}}{\mathbf{d_m}} \quad (31)$$

$d_A \stackrel{\text{def}}{=} d_m/(1 + z)$ is called the “angular-diameter distance”, so as to recover the classical relation $\phi = D/d_A$.

CosmoGravity also allows the inverse calculation: z as a function of ϕ

13 Photometry calculations

By considering the “wave surfaces” emitted by a source and reaching the observer, one shows that the observed brightness (power received per unit area normal to the direction of the source) E_0 of an object of intensity I_e (flux emitted per unit solid angle in the direction of

the observer) and cosmological shift z has (in the absence of absorption along the path) an observed brightness

$$E_0 = \frac{I_e}{d_m^2(1+z)^2} \quad (32)$$

Compared with the “classical” relation, there is a factor $(1+z)^2$ in the denominator. The explanation is that these photons undergo a spectral shift, hence a decrease of frequency ν and therefore of energy ($E = h\nu$) and, in addition, the “rate” (number per second) of photons is also divided by $dt_0/dt_e = 1+z$ (the photons emitted during a time dt_e are received during a time $dt_0 = (1+z)dt_e$, hence the second factor $(1+z)$ for the observed brightness (power received per unit orthogonal area)

If the object’s intensity is assumed isotropic, its “luminosity” (emitted power) is:

$$L_e = 4 \pi I_e \quad \text{and} \quad E_0 = \frac{L_e}{4 \pi d_m^2(1+z)^2} \stackrel{\text{d\'ef}}{=} \frac{L_e}{4 \pi d_L^2} \quad (33)$$

$d_L \stackrel{\text{d\'ef}}{=} d_m(1+z)$ is the “luminosity distance” (which allows one to recover the classical relation between brightness and distance, as was done above with the “angular-diameter distance”).

Magnitudes and distance moduli. Brightnesses and luminosities have an impressive dynamic range in astronomy (the Sun’s brightness is $\sim 10^{25}$ times that of the galaxies that the ELT telescope will be able to detect (in 2027?)). Since antiquity the concept of “magnitude” has been used, which was later defined under the name “apparent magnitude” m as a logarithmic scale of brightness E with $m \stackrel{\text{d\'ef}}{=} -2.5 \log_{10} E + \text{const.}$ And the “absolute magnitude” M is defined as the apparent magnitude the source would have if it were located at a distance of 10 pc; it is therefore a logarithmic measure of luminosity.

The “distance modulus” μ is defined as the difference $m - M$ for an object. One easily deduces that, in a transparent space,

$$\mu \stackrel{\text{d\'ef}}{=} m - M = -5 + 5 \log_{10} d_{\text{pc}} \quad (34)$$

Both m and M may be “bolometric” (all wavelengths and with a perfect detector) or depend on the spectral sensitivity curve of the instrumentation, but in all cases the distance modulus keeps the above expression (assuming perfect transparency of the medium between the source and the observer). If one observes “standard candles” one knows their luminosity (hence their M), and measuring their m gives their distance. Obviously, for cosmological distances it is the “luminosity distance” $d_L = d_m(1+z)$ that must be used in the classical relation:

$$\mu \stackrel{\text{d\'ef}}{=} m - M = -5 + 5 \log_{10} d_{\text{L(pc)}} \quad (35)$$

14 Single-fluid models

Analytic solutions exist for some of the Friedmann-Lemaître universe models. This is in particular (but not only) the case for those in which a single density parameter Ω_i is non-zero. These particular models also have the advantage of being approximations, at certain epochs, of a multi-fluid FL universe.

Since $\Omega_{\mathbf{m}}(\mathbf{t}) + \Omega_{\mathbf{r}}(\mathbf{t}) + \Omega_{\Lambda}(\mathbf{t}) + \Omega_{\mathbf{k}}(\mathbf{t}) = 1 \quad \forall \mathbf{t}$, if a single one of the Ω_i is non-zero it is therefore always equal to 1. Four cases arise: $\Omega_m(t) = 1$, $\Omega_r(t) = 1$, $\Omega_{\Lambda}(t) = 1$ and, why not, $\Omega_k(t) = 1$.

14.1 Matter, $\Omega_m = 1 \quad \forall t$

For this “dust” universe (non-relativistic matter), or Einstein-de Sitter universe (1932), $p \approx 0$ and: $\rho_m R^3 = \text{const} = \rho_{m0} R_0^3$

The Friedmann-Lemaître equation FL2 then implies (taking as time origin that of the singularity: $R(t=0) = 0$)

$$\mathbf{R}(\mathbf{t}) = (6\pi \mathbf{G} \rho_{m0} \mathbf{R}_0^3)^{\frac{1}{3}} \mathbf{t}^{\frac{2}{3}} \quad (36)$$

$$\Omega_m(t) \stackrel{\text{def}}{=} \frac{8\pi G \rho_m(t)}{3H^2(t)}, \Rightarrow \rho_{m0} = \frac{3H_0^2}{8\pi G} \quad (37)$$

Likewise, since

$$H \stackrel{\text{def}}{=} \frac{\dot{R}}{R}, \quad H(t) = \frac{2}{3t} \quad \text{and} \quad \mathbf{H}_0 = \frac{2}{3\mathbf{t}_0} \quad (38)$$

it follows

$$\mathbf{a}(\mathbf{t}) \stackrel{\text{def}}{=} \frac{\mathbf{R}(\mathbf{t})}{\mathbf{R}_0} = \left[\frac{3\mathbf{H}_0}{2} \right]^{2/3} \mathbf{t}^{\frac{2}{3}} \quad (39)$$

The particle horizon dp_o has a simple expression in this case. Indeed

$$E(x) = (1+x)^3 \quad \text{and} \quad \mathbf{d}_{p_o} = \frac{\mathbf{c}}{\mathbf{H}_0} \int_0^\infty \mathbf{E}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} = \frac{2\mathbf{c}}{\mathbf{H}_0} \quad (40)$$

and since

$$H = \frac{\dot{R}(t)}{R(t)} = \frac{2}{3t}, \quad (41)$$

then $d_{p_o} = 3ct_0$

14.2 Radiation, $\Omega_r = 1 \quad \forall t$

With $p_r = \frac{1}{3}\rho_r c^2$ and FL3: $\rho_r R^4 = \text{const} = \rho_{r0} R_0^4$

In this “Weinberg universe” the Friedmann-Lemaître equation FL2 implies:

$$R^2 \dot{R}^2 = \frac{8\pi G}{3} \rho_{r0} R_0^4 \Rightarrow \mathbf{R}(\mathbf{t}) = \left[\frac{32\pi \mathbf{G} \rho_{r0} \mathbf{R}_0^4}{3} \right]^{\frac{1}{4}} \mathbf{t}^{\frac{1}{2}} \quad (42)$$

$$\Omega_r(t) \stackrel{\text{def}}{=} \frac{8\pi G \rho_r(t)}{3H^2(t)}, \Rightarrow \rho_{r0} = \frac{3H_0^2}{8\pi G} \quad (43)$$

Likewise, since

$$H \stackrel{\text{def}}{=} \frac{\dot{R}}{R}, \quad H = \frac{1}{2t} \quad \text{and} \quad \mathbf{H}_0 = \frac{1}{2\mathbf{t}_0} \quad (44)$$

It follows

$$\mathbf{a}(\mathbf{t}) \stackrel{\text{def}}{=} \frac{\mathbf{R}(\mathbf{t})}{\mathbf{R}_0} = [2\mathbf{H}_0]^{\frac{1}{2}} \mathbf{t}^{\frac{1}{2}} \quad (45)$$

The particle horizon dp_o has, in this case as in the previous one, a simple expression:
 $dp_o = 2ct_0$

Note: Since the radiation energy density of a black body is

$$u_r = \rho_r c^2 = \frac{4\sigma T^4}{c} \quad \text{and that} \quad \Omega_r \stackrel{\text{def}}{=} \frac{8\pi G \rho_r}{3H^2} = 1 \quad (46)$$

the temperature T_r depends only on H :

$$T_r = \left[\frac{3H^2 c^3}{32\pi G \sigma} \right]^{\frac{1}{4}} \quad \text{with} \quad \sigma = \frac{2\pi^5 k^4}{15c^2 h^3} \quad \text{and} \quad \mathbf{T}_r = \left[\frac{45\mathbf{c}^5 \mathbf{h}^3}{64\pi^6 \mathbf{G} \mathbf{k}^4} \right]^{\frac{1}{4}} \mathbf{H}^{\frac{1}{2}} \quad (47)$$

14.3 Cosmological constant, $\Omega_\Lambda = 1 \quad \forall t$

The Friedmann-Lemaître equation FL2 implies, for this de-Sitter universe,

$$\frac{\dot{R}^2}{R^2} = \frac{\Lambda c^2}{3} = \text{const} \quad (48)$$

Therefore $\Lambda \geq 0$ and

$$\mathbf{H} = \pm \mathbf{c} \left[\frac{\Lambda}{3} \right]^{1/2} = \text{const} = \mathbf{H}_0 \quad (49)$$

and

$$\mathbf{a}(\mathbf{t}) \stackrel{\text{def}}{=} \frac{\mathbf{R}(\mathbf{t})}{\mathbf{R}_0} = \mathbf{e}^{\mathbf{H}_0 \cdot (\mathbf{t} - \mathbf{t}_0)} \quad (50)$$

14.4 Curvature, $\Omega_k = 1 \quad \forall t$

The sum of the Friedmann-Lemaître equations FL1+FL2 implies, for the Milne universe:
 $\ddot{R} = 0$. Therefore:

$$R(t) = \alpha t + \beta \quad , \quad \dot{R} = \alpha = \text{const} \quad \text{and} \quad H_0 \stackrel{\text{def}}{=} \frac{\dot{R}_0}{R_0} = \frac{\alpha}{R_0} \quad (51)$$

Consequently

$$\mathbf{a}(\mathbf{t}) \stackrel{\text{def}}{=} \frac{\mathbf{R}(\mathbf{t})}{\mathbf{R}_0} = \mathbf{H}_0 \mathbf{t} + \frac{\beta}{\mathbf{R}_0} = \mathbf{H}_0 \cdot \mathbf{t} + \text{const} \quad (52)$$

Taking as time origin that of $\mathbf{a}=0$

$$\mathbf{a}(\mathbf{t}) = \mathbf{H}_0 \cdot \mathbf{t} \quad \text{and} \quad \mathbf{t}_0 = (\mathbf{H}_0)^{-1} \quad (53)$$

15 Alternative cosmologies

15.1 Brief history

The Λ CDM model (with Λ a geometric constant) is, with different parameter values, the one published by G. Lemaître in 1931. However, it was the Einstein-de-Sitter (EdS) model of 1932 (without the Λ that Einstein had disowned) that was subsequently the most widely considered. In the 1980s, the measurements of the expansion rate and of the matter density, baryonic or not, with $\Omega_{m0} \approx 0.3$, gave an age for the EdS universe smaller than that of the oldest stars (13 billion years), which on the other hand was very well explained by $\Omega_\Lambda \approx 0.7$.

It was only in 1998 that the first measurements on SNIa-type supernovae confirmed a $\Omega_\Lambda > 0$, and that the balloon observations (“Boomerang” and “Maxima”) (1999-2000) revealed the apparent diameter of the first acoustic peak of the cosmic microwave background. By combining the observational constraints derived from the brightness-redshift relations of SNIa supernovae with those of the apparent diameter ($\phi \approx 1^\circ$) of the first acoustic peak of the CMB, the Λ CDM model with $\Omega_{m0} \approx 0.3$ and $\Omega_{\Lambda 0} \approx 0.7$, giving an age of ~ 13.8 billion years, became the standard model of cosmology, still in use in 2024.

15.2 Tensions

But new measurements have revealed “tensions” with the standard Λ CDM model. Among them:

- **“ H_0 tension”** : direct measurements from “standard candles” give $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, whereas those derived more indirectly from the CMB converge towards $H_0 \approx 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with statistical discrepancies of several σ .
- **“ σ_8 tension”**: the observed structuring of the universe on a scale of about¹⁹ $8h^{-1} \text{ Mpc}$ is significantly weaker than that inferred from the CMB anisotropies.
- **“Dipole tension”** : the observed dipole component of the CMB, interpreted as the Doppler-Fizeau effect of our genuine motion ($369 \pm 1 \text{ km/s}$) with respect to this radiation ($z \sim 1000$), is close to but significantly different, in magnitude and orientation, from the one observed at “smaller” distance on the quasars ($z \sim 2$).

15.3 New theories

In parallel, new cosmological theories have been proposed

- Inflation: these theories do not necessarily challenge the Λ CDM model but could explain the “initial” values (before $\sim 10^{-30} \text{ s}$) of its parameters
- Dark energy
- and many others...

CosmoGravity now includes simulations of universes with “dark energy” as an alternative to the geometric cosmological constant Λ .

¹⁹ $h \stackrel{\text{def}}{=} H_0(\text{km s}^{-1}\text{Mpc}^{-1})/100$

16 Dark energy

16.1 Relativistic equations of state

Matter and radiation enter the energy-momentum tensor of a perfect fluid only through their “equation of state”: $p = p(\rho)$. One can thus, while remaining within the framework of the Friedmann-Lemaître equations, take into account different “fluids” i , known or hypothetical, characterised by an equation of state of the type $p_i = w_i \rho_i c^2$. The total mass density and pressure of a universe with n “fluids” are then: $\rho = \sum_{i=1}^n \rho_i$ and $p = c^2 \sum_{i=1}^n w_i \rho_i$.

For “dust” (Dust d) (non-relativistic matter) $p \ll \rho c^2$ and $w_d = w_{nr} \approx 0$. For radiation (light and ultra-relativistic particles) $w_r = 1/3$.

The evolution of the scale factor a and that of the other cosmological parameters can be generalised to a mixture of n fluids i . For the evolution of the expansion rate $H = \dot{a}/a$ one obtains:

$$\frac{H}{H_0} = \left[\sum_{i=1}^n \Omega_i a^{-3(1+w_i)} \right]^{\frac{1}{2}} = \left[\sum_{i=1}^n \Omega_i (1+z)^{3(1+w_i)} \right]^{\frac{1}{2}} \quad (54)$$

The curvature of space is then equivalent to a fluid with $w_k = -1/3$.

16.2 From Λ to dark energy

The cosmological constant Λ is equivalent to (and can be interpreted as²⁰) a “vacuum” fluid, Lorentz-invariant, with $w_\Lambda = w_v = -1$ and mass density $\rho_{DE} = \frac{\Lambda c^2}{8\Pi G}$. This substitution indeed leaves mathematically unchanged Einstein’s 1917 equation (4) and consequently those of Friedmann-Lemaître, provided the cosmological principle is retained.

Moreover, the geometric cosmological constant Λ can pose a severe fine-tuning problem²¹ and attempts have been made to replace it with a physical fluid. If this new, for now speculative, entity is described by an equation of state with parameter w , this parameter must be spatially constant (cosmological principle) but different from -1, or even variable in time t (that is, with z and a).

The geometric character of Λ can also be a (philosophical) motivation for replacing it with a physical entity. It should be noted, however, that in Einstein’s 1917 field equation (4), which includes Λ and has a far greater mathematical generality, while one may “add” new physical fluids on the left-hand side of the equation, “replacing” the Λ term by a physical fluid on the right-hand side is a modification of the mathematical equation (4), unless this DE (Dark Energy) fluid has parameter $w_{DE} = \text{const} = -1$ and mass density $\rho_{DE} = \frac{\Lambda c^2}{8\Pi G}$.

One sometimes distinguishes “dark energy” ($w_{DE}(z) \neq -1$), “quintessence” ($w_Q \neq \text{const}$), “phantom energy” ($w_{PE} < -1$) ... but it has become usual and simpler to generalise the term

²⁰Lemaître, 1934, Proc.National. Acad. Sciences USA, vol 20, pp12-17

²¹its very small but constant value indeed makes it insignificant in the primordial universe, but if its value were larger, the early acceleration of the expansion that it would have caused would have prevented the formation of structures.

“dark energy” to the whole set of possibilities and to use the CPL parametrisation²² with two (constant) parameters w_0 and w_1 :

$$w(z) = w_0 + w_1 \frac{z}{1+z} = w_0 + w_1(1-a) \quad (55)$$

In this representation Λ appears as a particular case of a dark energy with parameters $w_0 = -1$ and $w_1 = 0$

Observations make it possible to constrain the region of our universe in a (w_0, w_1) plane, as is done for the $(\Omega_{m0}, \Omega_{\Lambda0})$ plane.

The more-than-a-century-old²³ geometric cosmological constant Λ , or its physical equivalent ($w_0 = -1$ and $w_1 = 0$), is, in early 2025, seriously challenged by a dark energy through observations such as the ongoing ones from DESI (Dark Energy Spectroscopic Instrument), installed on the 4 m (and 1970) Kitt Peak telescope, which allows the simultaneous acquisition of 5000 spectra in a field of 8 square degrees. To be continued ...

16.3 Calculations

As with $E(x)$ for the model with the constant Λ , some functions simplify the writing of the relations, $Y(x)$ and $F(x)$:

$$\mathbf{Y}(\mathbf{x}) \stackrel{\text{def}}{=} \exp \{ -3(\mathbf{1} + \mathbf{w}_0 + \mathbf{w}_1) \log \mathbf{x} - 3\mathbf{w}_1(\mathbf{1} - \mathbf{x}) \} \quad (56)$$

$$\mathbf{F}(\mathbf{x}) \stackrel{\text{def}}{=} \left[\frac{\mathbf{H}(\mathbf{x})}{\mathbf{H}_0} \right]^2 = (\mathbf{1} + \mathbf{x})^2 \Omega_{k0} + (\mathbf{1} + \mathbf{x})^3 \Omega_{m0} + (\mathbf{1} + \mathbf{x})^4 \Omega_{r0} + \mathbf{Y}((\mathbf{1} + \mathbf{x})^{-1}) \Omega_{DE0} \quad (57)$$

In this way the first and second derivatives of $a(\tau)$ become:

$$\frac{d\mathbf{a}}{d\tau} = \left[\frac{\Omega_{r0}}{\mathbf{a}^2} + \frac{\Omega_{m0}}{\mathbf{a}} + \Omega_{DE0} \mathbf{a}^2 \mathbf{Y}(\mathbf{a}) + \Omega_{k0} \right]^{\frac{1}{2}} \quad (58)$$

$$\frac{d^2\mathbf{a}}{d\tau^2} = -\frac{\Omega_{r0}}{\mathbf{a}^3} - \frac{1}{2} \frac{\Omega_{m0}}{\mathbf{a}^2} + \Omega_{DE0} \left[\mathbf{a} \mathbf{Y}(\mathbf{a}) + \frac{\mathbf{a}^2}{2} \frac{d\mathbf{Y}}{d\mathbf{a}} \right] \quad (59)$$

and that of the metric distance, according to negative, zero or positive curvature:

$$d_{p0-} = \frac{c}{H_0 |\Omega_{k0}|^{\frac{1}{2}}} \sinh \left\{ |\Omega_{k0}|^{\frac{1}{2}} \int_0^\infty \mathbf{F}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \right\} \quad (60)$$

$$d_{p0} = \frac{c}{H_0} \int_0^\infty \mathbf{F}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \quad (61)$$

$$d_{p0+} = \frac{c}{H_0 |\Omega_{k0}|^{\frac{1}{2}}} \sin \left\{ |\Omega_{k0}|^{\frac{1}{2}} \int_0^\infty \mathbf{F}^{-\frac{1}{2}}(\mathbf{x}) d\mathbf{x} \right\} \quad (62)$$

²²Chevalier & Polarski 2001, Int. J. Mod. Phys. D10,213; Linder 2003 Phys. Rev. Lett. 90,091301 and Sahni & Starobinsky, Int. J. Mod. Phys., 2006, Vol. 15, pp 2105-2132 or arXiv:astro-ph/0610026

²³Einstein, 1917, Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin), Seite 142-152

The remaining expressions used in the calculations of E^{24} , d_L , d_A , θ , d_{LT} , $l \dots$ are identical to those of the model with Λ , replacing $E(x)^{25}$ by $F(x)$.

Introducing terms for other fluids in the expressions of the metric distance d_m opens the way to observational tests and to constraining the new parameters, by inverting the relation $\{\text{parameters}\} \rightarrow \{\text{observables}\}$, since the latter, such as the brightness or the apparent diameter, depend on d_m^{26} .

With dark energy, if one defines arbitrarily (N for normalised):

$$\rho_{DEN} \stackrel{def}{=} \rho_{DE0} Y(a) \quad \text{and} \quad \Omega_{DEN} \stackrel{def}{=} \frac{8\pi G \rho_{DEN}}{3H^2} = \Omega_{DE0} Y(a) \quad (63)$$

one recovers the closure equation (2) **TO BE CHECKED:**

$$\Omega_r(t) + \Omega_m(t) + \Omega_{DEN}(t) + \Omega_k(t) = 1 \quad \forall t \quad (64)$$

16.4 The “Big Rip” and the “Big Fall”

The standard Λ CDM model begins with a “Big Bang” and its future is an infinite exponential growth of the scale factor. Because of this large-scale expansion and the progressive extinction of the stars, a great progressive cooling lies ahead (*Big Chill* or *Big Freeze*).

The present observational constraints (early 2025) on our universe in the (w_0, w_1) plane are compatible, at less than 2σ , with $w_0 = -1$ and $w_1 = 0$ (the geometric cosmological constant Λ), but they are also compatible with slightly different values of w_0 and w_1 , some of which predict a “**Big Rip**”.

Unlike the “Big Chill” (or “Big Freeze”) of the standard Λ CDM model, where the expansion rate tends towards a constant (and smaller than its present value in the standard Λ CDM model) that preserves the structures not currently expanding, dark energy can allow a “Big Rip” (great tearing-apart), where $a(t)$ and the expansion rate H itself tend towards infinity in one and the same finite time and then disaggregate, faster and faster, down to the smallest material structures. For matter this is therefore a (physical) singularity, existentially as extreme as that of the $a = 0$ of the “Big Bang”.

And with dark energy it is also possible to conceive of universes without a Big Bang or a Big Crunch but with a “Big Fall” and a “Big Rip”, that is, with a finite past and future between 2 vertical asymptotes of $a(t)$, and hence a finite duration of this Universe²⁷.

For the simulations with the dark-energy option (and the CPL parametrisation), Cosmo-Gravity first checks whether there is a “Big Crunch”. If (and only if) the answer is negative, there remain (at least) 4 possibilities:

1. If $w_1 < 0$ then $w \rightarrow +\infty$ when $z \rightarrow -1$ no Big Rip.

²⁴photometric brightness

²⁵the function $E(x)$

²⁶Obviously, increasing the number of free parameters of the model with the same data widens the uncertainties on each of them in the inversion

²⁷Such universes have no $a(t)$ close to 0 and therefore no nucleosynthesis phase for the light elements.

2. If $w_1 > 0$ then $w \rightarrow -\infty$ when $z \rightarrow -1$ Big Rip (except for an *oscillating universe* between two non-zero values of the scale factor)
3. If $w_1 = 0$ and $w_0 < -1$ then $w(\text{constant}) < -1$ Big Rip
4. If $w_1 = 0$ and $w_0 > -1$ then $w(\text{constant}) > -1$ no Big Rip

In cases 2 and 3 above of Big Rip, the time $t_{BR} - t_0$ remaining before the Big Rip is then

$$t_{BR} - t_0 = \frac{1}{H_0} \int_{-1}^0 (1+x)^{-1} F^{-\frac{1}{2}}(x) dx \quad (65)$$

The “duration” of this universe is obviously $t_0 + t_{BR}$.

17 Universes with a non-monotonic $a(t)$ function

Universes with:

- a Big Bang and a Big Crunch
- neither a Big Bang nor a Big Crunch
- a Big Fall and a Big Rip

present a maximum or a minimum of $a(t)$, and therefore respectively a minimum or a maximum of the observed z . And if one chooses an observable z (and hence an $a(t)$), there correspond to it 2 possibilities for the times and distances: one possibility while z is increasing and one while it is decreasing. The choice of this increase or decrease can (with a little patience) be regarded as an observable (cf. §11) and, in these cases, CosmoGravity leaves the user the choice of increasing or decreasing z to compute distances and times.

17.1 maximum of $a(t)$ and hence minimum z_{min} of z

Metric distances: Compared with §9, the integration of the function $E^{\frac{1}{2}}(x)$ is

- from 0 to z for increasing z
- the sum from 0 to z_{min} and from z_{min} to z for decreasing z .

Particle horizon

- integration from 0 to ∞

Event horizon

- sum of the integrations from z_{min} to 0 and from z_{min} to ∞

17.2 minimum of $a(t)$ and hence maximum z_{max} of z

Metric distances: Compared with §9, the integration of the function $E^{\frac{1}{2}}(x)$ is

- from 0 to z for decreasing z
- the sum from 0 to z_{max} and from z_{max} to z for increasing z .

Event horizon

- integration from 0 to ∞

Particle horizon

- sum of the integrations from 0 to z_{max} and from z_{max} to ∞

18 From UNIVERS to TRAJECTOIRES: Structuring

CosmoGravity allows simulations of a universe satisfying the cosmological principle, that is, with a homogeneous and isotropic distribution of the content. For our present universe one can say that what is homogeneous is its distribution of ... galaxy superclusters (10^5 galaxies each).

Simulating the formation of galaxy superclusters and of ever smaller, more diverse and more numerous structures requires ever more computing power (and time): the growth of the overdensities is no longer linear and the simulation requires a “tiling” into a number N of contiguous tiles that quickly becomes ... astronomical. Hence the need for a computing power (and duration) beyond the reach of the *client server* CosmoGravity.

One may note, however, that for smaller structures the transformation is relatively very rapid: going from a cloud of gas and dust of about 1 ly to an open cluster of a few tens of stars takes $\sim 10^5$ years, i.e. $\sim 10^{-5}$ times the age of the universe.

And the present level of structuring of the universe on small scales is extreme: in our near environment alone, in the disc of our galaxy, the distance to the nearest star is ≈ 4 light-years (i.e. $\sim 4 \cdot 10^{16}$ m). The mean mass density of our planet is $\rho_{\oplus} \approx 5500 \text{ kg} \cdot \text{m}^3$, that of our star $\rho_{\odot} \approx 1400 \text{ kg} \cdot \text{m}^{-3}$.

Since the diameter of the Sun is $D_{\odot} \approx 1.4 \cdot 10^9$ m, the ratio of distances between $D_{\odot}$ and the distance to the nearest star is $\sim 3 \cdot 10^7$... and the ratio of volumes $\sim 10^{23}$: in our near environment the content is $\sim 10^{23} \text{ m}^3$ of vacuum for 1 m^3 of celestial body.

Because of this extreme contrast, the geometry of space-time **in the vicinity of a single isolated object** is thus essentially defined by the mass of that object. One thereby recovers a high degree of symmetry which allows, with the same general relativity and the “metric” it associates with the vicinity of that object, dynamic simulations that are economical in computing power and time.

And this is the aim of the TRAJECTOIRES simulations of CosmoGravity: they are dynamic, showing the motion of space travellers (they and their vehicles being assumed of negligible mass compared with that of the object) and displaying in particular the different flows of time for the travellers and for a distant observer, the felt constraints, ...